

CoronaryTwin

A 3D coronary "twin" that shows **where** risk sits (overall CAD and the LAD, LCX and RCA arteries), **how sure** the model is, **why** it thinks so, and **what would change it**, from 51 routine clinical measurements.

Decision support and educational use only. Not a diagnosis and not a substitute for formal diagnostic imaging or clinical judgment. The 3D coloring is schematic: it comes from vessel-level estimates, not lesion locations.

Results at a glance

Every number below comes from patients the model never saw (nested 5×5 repeated cross-validation, no tuning on test folds). Brackets are 95% confidence intervals.

Overall CAD · ROC-AUC 0.92 0.91–0.93 · logistic regression	LAD · ROC-AUC 0.85 0.83–0.87 · random forest	LCX · ROC-AUC 0.72 0.70–0.75 · random forest	RCA · ROC-AUC 0.72 0.69–0.74 · logistic regression
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Conformal coverage **90.5–90.7%** (target 90%). When both outcomes remain possible, the artery is drawn **Uncertain** (translucent, dashed) instead of looking certain.

What the app does

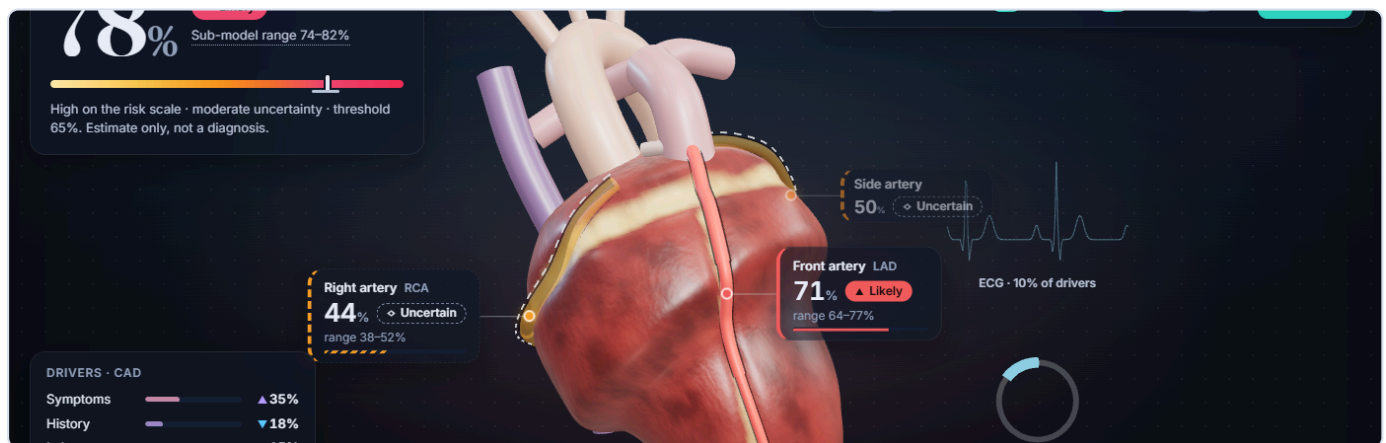
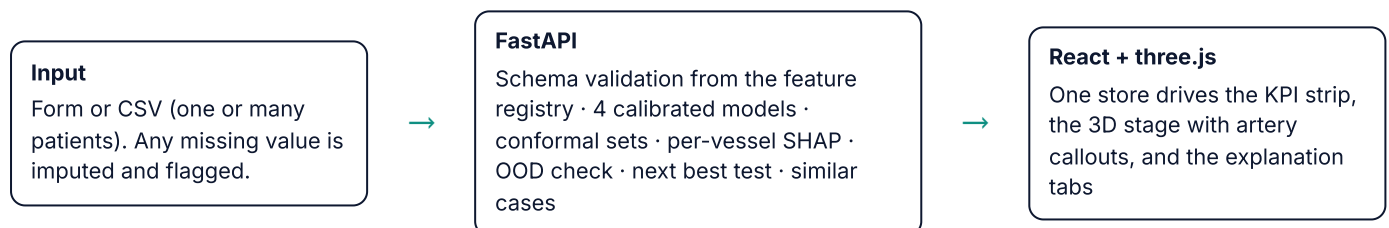


Fig. 1 · The 3D stage: overall estimate, a callout on each artery, and the drivers card. LCX and RCA are Uncertain (dashed).

What makes it different

- **Uncertainty you can see:** calibrated probabilities, conformal "Uncertain" states, and a sub-model range on every estimate.
- **Explanations on the anatomy:** SHAP per artery, a waterfall that adds up exactly, and glows on the heart linked to the chart.
- **Clinically shaped tools:** next best test (which missing value matters most), similar historical cases, out-of-distribution warning, cohort view.
- **Honest by construction:** safety banner in every view, captions generated from the patient's own numbers, and the limits stated in the app.

1 · Data, preprocessing and the leakage guard

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Dataset: Extension of Z-Alizadeh Sani (UCI id 411, DOI 10.24432/C5461K). 303 catheterized patients from one center, 59 columns, no missing values.

TARGET	DEFINITION	POSITIVE
CAD	Overall coronary artery disease (Cath)	71.3%
LAD	Left anterior descending $\geq 50\%$ stenosis	58.4%
LCX	Left circumflex $\geq 50\%$ stenosis	39.3%
RCA	Right coronary artery $\geq 50\%$ stenosis	37.6%

Inputs: 51 features in 7 explanation groups: Demographics, History, Symptoms, Vitals, Labs, ECG and Echo. Excluded with reasons: the four label columns, Weight and Length (BMI is computed exactly from them), Obesity (equals $\text{BMI} > 25$) and Exertional CP (constant).

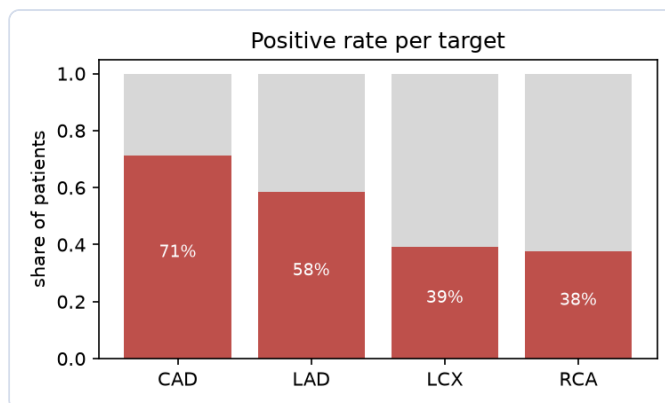


Fig. 2 · Class balance per target.

Feature registry: one source of truth

`features.yaml` holds each input's name, source column, encoding, group, valid and normal range, whether it is modifiable, and patient wording. It generates the API's input validation, the form, the SHAP groups, the what-if sliders and the patient text. A new feature needs no code change, and a test proves it.

Leakage guard

- Model inputs are built from a **whitelist** (the registry), and a **blacklist** blocks every label column. `test_leakage.py` fails if one reaches `fit`.
- All preprocessing (imputation, scaling, one-hot) is an unfitted pipeline that is **fitted inside each training fold only**.
- Audit: no single input has a ROC-AUC near 1.0 (highest: typical chest pain, 0.80). Overall CAD equals "any stenotic vessel" for 302 of 303 patients, which is consistent.

2 · Models and validation

- Four families** per target (logistic regression, random forest, XGBoost, LightGBM), tuned inside an **inner** 3-fold search and scored on an **outer** 5-fold $\times$ 5-repeat split. The simplest family within one standard error of the best is kept.
- Calibration:** Platt scaling with 5 cross-fitted copies; their spread gives the **sub-model range** shown for every estimate (model disagreement, not a CI).
- Conformal prediction** at $\alpha = 0.10$ turns each probability into Likely / Unlikely / Uncertain, with coverage checked on held-out folds.
- Thresholds** favor recall (≥ 0.85), because a missed narrowing costs more than a false alarm.

3 · Results (nested cross-validation, out-of-fold)

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TARGET	MODEL	ROC-AUC (95% CI)	BRIER	THRESHOLD	RECALL	PRECISION	F1	COVERAGE	UNCERTAIN
CAD	Logistic regression	0.92 (0.91–0.93)	0.104	0.65	0.85	0.92	0.89	90.5%	11%
LAD	Random forest	0.85 (0.83–0.87)	0.158	0.50	0.85	0.79	0.82	90.6%	35%
LCX	Random forest	0.72 (0.70–0.75)	0.206	0.29	0.86	0.50	0.63	90.5%	58%
RCA	Logistic regression	0.72 (0.69–0.74)	0.204	0.28	0.85	0.48	0.62	90.7%	55%

LCX and RCA are harder to predict from routine data. Their learning curves still rise at the full training size, and these models say **Uncertain** for more patients by design rather than guessing.

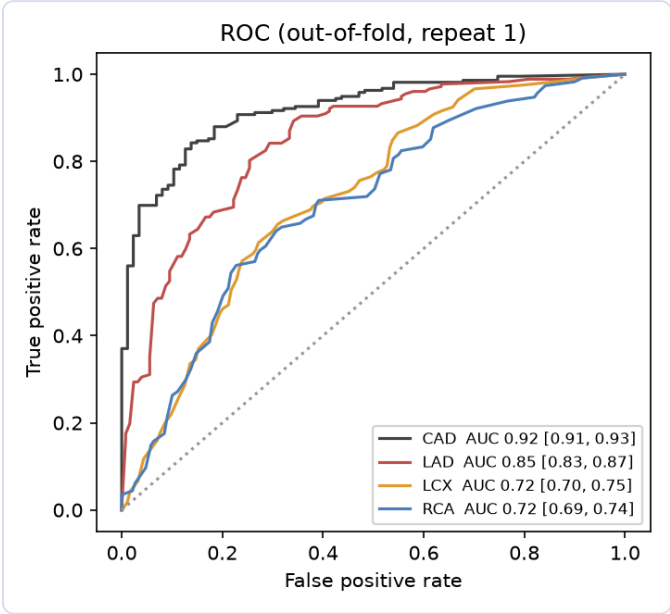


Fig. 3 · Out-of-fold ROC curves.

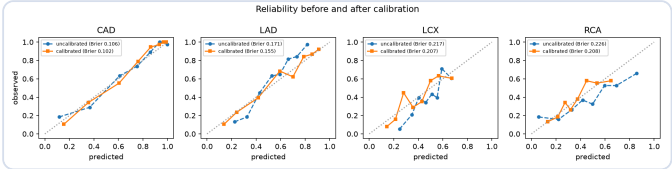


Fig. 4 · Reliability after Platt scaling.

Checks beyond the headline numbers

EXPERIMENT	RESULT
Paired bootstrap, every family vs the selected one	No family is reliably better (every 95% CI of Δ AUC includes 0).
Soft-vote ensemble of all four families	AUC 0.923 / 0.850 / 0.719 / 0.732: no meaningful gain, so it was not adopted.
Classifier chain (overall CAD estimate as an input to the artery models, nested)	Δ AUC +0.006 / +0.001 / -0.007 (LAD/LCX/RCA), no reliable gain.
Ablation by measurement group	Symptoms help every model most (-0.03 to -0.07 AUC when removed); Echo matters for LAD (-0.037).
Engineered features (risk-factor count, ECG-abnormality count, LDL/HDL, interactions)	No reliable gain; reported, not used.
Input noise of 10% of each feature's spread	Mean $ \Delta p \leq 0.03$; 0% Likely→Unlikely flips for all four models.
Decision curves	Net benefit above "treat all" and "treat none" across clinically used thresholds.
Subgroups (sex, age bands)	Reported in the app; groups too small to judge are marked.

All of these are reproducible with `python tasks.py experiments` and appear in the app's Model-trust tab.

4 · Explainability

- **SHAP per artery:** each of the four models has its own explanation (LinearExplainer or TreeExplainer). Selecting the LAD shows LAD drivers.
- **Waterfall:** from the average patient to this patient's model score, one bar per measurement group, in the model's own units so it adds up exactly. Hovering a group lights up its glow or gauge on the 3D heart.
- **Stability:** feature rankings across 5 refits agree with Spearman ρ 0.78–0.92.
- **LIME cross-check:** on 20 patients per model, LIME and SHAP share 60–70% of their top-3 drivers (chance about 6%).
- **Two audiences:** a clinician view with values and units, and a plain-language patient view with summary cards.

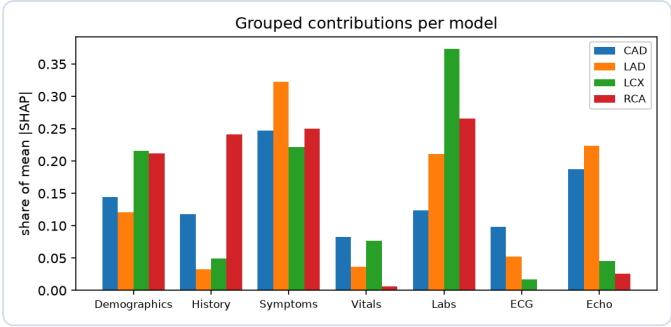


Fig. 5 · Share of each measurement group in each model's explanation.

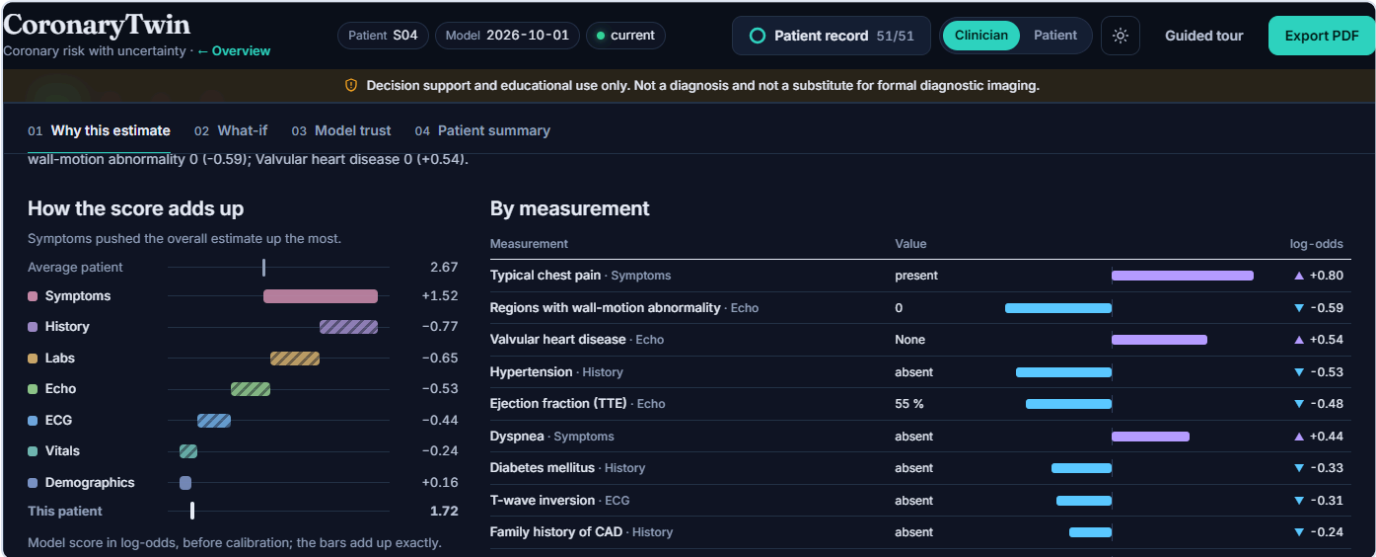


Fig. 6 · "Why this estimate": waterfall, per-measurement drivers, and similar historical cases.

From explanation to action, honestly

TOOL	WHAT IT DOES	EVIDENCE
What-if	Moves modifiable values; arteries recolor live; the counterfactual finds the smallest change below the threshold.	Labeled model sensitivity, not advice. Explains when changes are small (strongest drivers not modifiable).
Next best test	For missing values, ranks the test that would change the estimate most, using values from similar patients.	Picked the single most informative test in 68% of 60 patients (chance 33%).
Similar cases	The 5 nearest training patients in a SHAP-weighted space, as coarse summaries with their outcomes.	No raw rows or row numbers are ever returned.
Out-of-distribution warning	Flags values beyond the training range and unusual combinations (Mahalanobis).	1.3% false alarms on real patients; 100% of out-of-range synthetic patients flagged.

5 · 3D visualization

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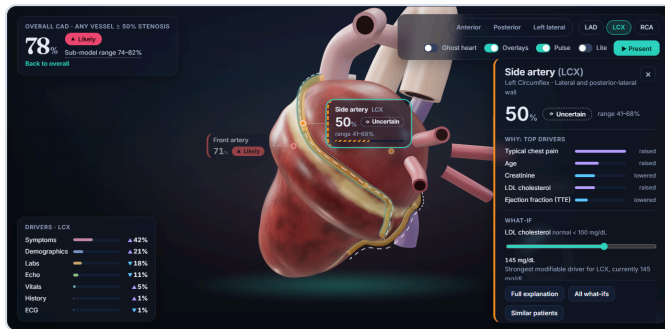


Fig. 7 · LCX selected: the camera flies in and the detail panel shows its drivers and a what-if slider.



Fig. 8 · Posterior view in ghost mode: the arteries' full course is visible.

- **Procedural anatomy, no third-party meshes:** one analytic surface with ventricles, atria and appendages, and grooves fitted to the artery paths. Arteries are projected exactly onto it, so they never float off the heart (tested). Muscle, epicardial fat, the great vessels and a wet sheen are generated in code.
- **Encoding:** color = calibrated probability on a yellow → rose ramp used only for risk (luminance falls monotonically, tested). Uncertainty = translucency, a dashed outline and a slow, irregular pulse. Color is always paired with a number and a word.
- **Interaction:** rotate, zoom, presets, click an artery (the camera flies in and a detail panel opens with its drivers and a what-if slider), ghost heart, explanation overlays, cohort averages, and **Present mode**, a captioned walkthrough generated from the patient's numbers.
- **Performance:** about 110k triangles, renders on demand. Measured on **Intel HD Graphics 620** (integrated): 36–44 fps with all effects, 60 fps in Lite mode, which switches on automatically below 14 fps. A 2D SVG fallback covers browsers without WebGL.

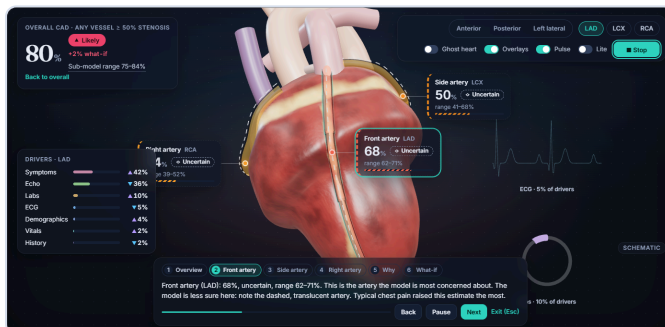


Fig. 9 · Present mode: steps, a data-driven caption, and the camera on the LAD.

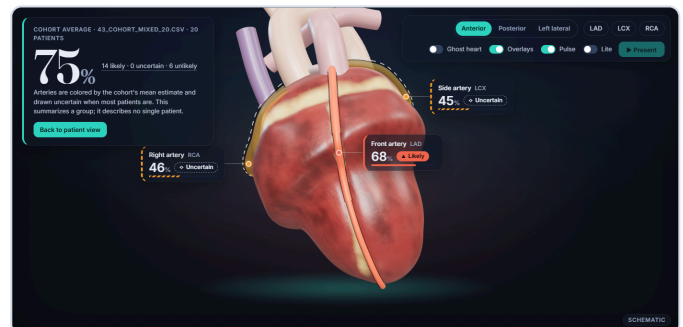


Fig. 10 · Cohort mode: a 20-patient upload shown as averages on the heart, clearly badged.

6 · Integration, running it, limits and credits

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Run it

- **Docker:** `docker compose up --build → http://localhost:8080` (API docs at `:8000/docs`).
- **Local:** `python tasks.py setup`, `python tasks.py serve`, `python tasks.py web`.
- **Try it:** Watch the 90-second tour on the landing page, or upload a file from `csv/` (50 synthetic samples, single and cohort).
- **Rebuild everything:** `data → train → explain → experiments`; seeds are fixed, so numbers reproduce.

Engineering

- 120 Python tests (leakage, API contract, SHAP additivity, registries, specialties) and 84 web tests, plus an end-to-end browser test of the judge path.
- CI: registry check, tests, web build, both Docker images, a smoke test through nginx, and the end-to-end test.
- Registries (`features.yaml`, `registry.json`, `anatomy.json`) are cross-checked; a new feature, model or artery is a registry edit.
- Privacy: patient inputs and IDs are never stored; a SQLite log keeps anonymous outputs only (`view_db.py`).

Safety layers

- Persistent banner in every view, in Present mode and in the exported PDF; first-use acknowledgement.
- "Disclaimer" field in every API response; wording says "estimated" and "likely", never "diagnosed".
- Schematic tag on the 3D stage; captions never say more than the numbers.

Limitations

- 303 patients from a single center; **no external validation**. The original and extended Z-Alizadeh Sani files contain the same patients.
- Labels are per vessel ($\geq 50\%$ stenosis), not per lesion: the 3D coloring cannot show where a narrowing is.
- Explanations describe associations the model learned, not biological cause; what-if is model sensitivity, not treatment advice.
- LCX and RCA discrimination is moderate (AUC ≈ 0.72).

Future work

- External validation on an independent cohort; a browser-only (ONNX) inference mode; a reviewed regional-language patient view.

Credits

Data: Alizadehsani R., Roshanzamir M., Sani Z. (2013). Extension of Z-Alizadeh Sani dataset. UCI Machine Learning Repository, DOI 10.24432/C5461K. 3D: all geometry and textures procedural (no third-party assets). Libraries: scikit-learn, XGBoost, LightGBM, SHAP, LIME, FastAPI, React, three.js / React Three Fiber, under their open-source licenses.

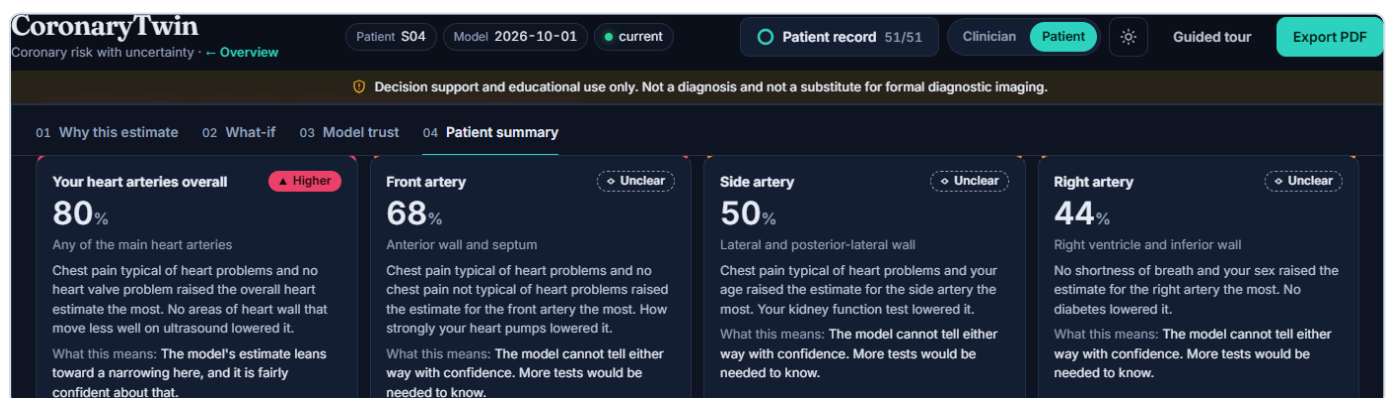


Fig. 11 · Patient view: plain-language cards and "Please talk to your doctor about these results."